

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

School of Computer Engineering

PRACTICAL TRAINING REPORT

ON

PwC Advisory Launchpad Program

CEDA Track: Data Engineering

Industry Related Training & Certification Program

Submitted By

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Faculty Guide

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STUDENT DECLARATION

I, **Sanidhya Kamthan** (Roll No. **23052909**), a student of the **School of Computer Engineering, Kalinga Institute of Industrial Technology (KIIT)**, hereby declare that this Practical Training Report titled **"PwC Advisory Launchpad Program, CEDA Track: Data Engineering"** has been prepared by me as a part of the Practical Training requirement for the Bachelor of Technology (B.Tech.) programme.

This report is based on the knowledge, practical exercises, assessments and learning activities completed by me during the PwC Advisory Launchpad Program conducted through the PwC Launchpad learning platform.

The training was undertaken during the period from **February 2026 to July 2026**. During this period, I completed structured learning modules covering programming fundamentals using Python, data engineering, modern data systems, data analytics, soft skills, generative AI and prompt engineering.

The contents of this report represent my understanding of the topics covered during the training and the practical activities completed as part of the programme. I have prepared this report sincerely for academic evaluation, and to the best of my knowledge, the information presented is based on my actual training experience and learning activities.

Place: Bhubaneswar

Date: 21/08/2026

Student Signature: 

Sanidhya Kamthan

Roll No.: **23052909**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Kalinga Institute of Industrial Technology (KIIT)** for providing me with the opportunity to undertake this Industry Related Training and Certification Program as part of my Bachelor of Technology curriculum.

I am thankful to my faculty guide, **Sayan Dutta**, for providing guidance and support during the preparation of this training report. The opportunity to document and reflect on the training has helped me understand how the concepts covered during the programme relate to my academic learning and future professional requirements.

I would also like to express my gratitude to **PwC Acceleration Centers in India** for providing the Advisory Launchpad learning programme. The programme offered structured learning across programming, data engineering, modern data systems, data analytics, generative AI and professional skills.

The self-paced modules, practical exercises, quizzes and assessments provided me with an opportunity to learn concepts systematically and apply them through problem-based activities. The programme also helped me understand the importance of continuous learning and adapting to emerging technologies.

Finally, I would like to thank everyone who directly or indirectly supported me during the completion of the training and the preparation of this report.

Sanidhya Kamthan

Roll No.: **23052909**

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INTRODUCTION

The technology industry is continuously evolving with the adoption of data-driven systems, cloud platforms, artificial intelligence and modern software technologies. For a Computer Science student, understanding programming and theoretical concepts is important, but practical exposure is equally necessary to understand how these concepts are applied to real-world problems.

With the objective of strengthening my technical and professional foundation, I participated in the **PwC Advisory Launchpad Program under the CEDA Track**. The programme was designed as an industry-oriented learning experience covering areas such as programming fundamentals, data engineering, modern data systems, data analytics, generative AI and professional skills.

The training followed a structured learning approach through the PwC Launchpad learning platform. It included self-paced modules, quizzes, programming exercises, practical activities and assessments. Instead of focusing only on theoretical concepts, several modules required solving practical problems and applying the concepts through coding and data-based exercises.

One of the major technical components of the programme was **Programming Fundamentals using Python**. The training covered Python basics, control structures, collections, functions, file handling, classes and objects, inheritance, polymorphism, regular expressions and exception handling. I also completed practical programming exercises based on these concepts.

The **Data Engineering** component introduced me to data manipulation and analysis using Pandas, exploratory data analysis, descriptive statistics, missing-value handling, correlation analysis, visualization, web scraping and basic machine learning applications.

The **Modern Data Systems** component provided practical exposure to SQL operations, database concepts, NoSQL and MongoDB, CRUD operations, indexing and analytical reporting. The programme also included **Generative AI and Prompt Engineering**, covering generative models, transformers, attention mechanisms, prompt design, AI agents and industry applications of GenAI.

Along with technical learning, the programme included **Soft Skills** modules covering communication, active listening, non-verbal communication, presentation skills, questioning techniques, time management, remote working and cultural sensitization.

Overall, the training provided a broad industry-oriented learning experience and helped me connect my academic knowledge with practical problems involving programming, data and emerging technologies.

ABOUT PwC ADVISORY LAUNCHPAD PROGRAM

The **PwC Advisory Launchpad Program** is an industry-oriented learning and skill development programme offered through PwC Acceleration Centers in India. The programme is designed to help students develop technical and professional capabilities that are relevant to modern technology and advisory environments.

The programme combines structured digital learning with practical exercises and assessments. Learners progress through different modules and activities while tracking their progress through the Launchpad platform.

I participated in the **CEDA Track, Data Engineering**, which focused on developing a foundation in programming, data engineering, modern data systems, data analytics and emerging technologies.

The training platform provided learning activities in different formats, including videos, quizzes and practical programming exercises. Many modules followed a learning sequence where concepts were introduced through learning material and then reinforced through practical exercises and assessments.

The programme also used an **XP-based progress system and leaderboard**, which provided a way to track individual learning progress. According to my training dashboard, I completed the required activities and reached **Level 10 with 3,931 XP**. My dashboard also showed **100% completion** of the learning path.

The major learning areas covered during my training were:

1. **Programming Fundamentals using Python**
2. **Data Engineering**
3. **Modern Data Systems**
4. **Generative AI and Prompt Engineering**
5. **Soft Skills**

The programme therefore provided exposure to both core technical foundations and emerging areas of technology. This combination helped me understand how programming and data skills can contribute to larger technology and business problems.

TRAINING OBJECTIVES

The primary objectives of undertaking the PwC Advisory Launchpad Program under the CEDA Track were as follows:

1. **Strengthen Programming Fundamentals**
   Develop a strong foundation in Python programming, including syntax, control structures, collections, functions and modules.

2. **Develop Data Engineering Skills**
   Gain practical understanding of data processing, Pandas, data manipulation and exploratory data analysis.

3. **Understand Modern Data Systems**
   Learn SQL operations, database concepts, CRUD operations, indexing and NoSQL databases such as MongoDB.

4. **Build Generative AI Knowledge**
   Understand Generative AI, generative models, Transformers, attention mechanisms and modern AI applications.

5. **Develop Prompt Engineering & AI Skills**
   Practice designing effective prompts and gain introductory exposure to AI agents and their workflows.

6. **Improve Professional & Industry Skills**
   Strengthen communication, presentation, active listening, time management and self-learning abilities.

TRAINING DETAILS

The following table provides the details of the Industry Related Training undertaken through the **PwC Advisory Launchpad Program**.

Particular	Details
Training Title	PwC Advisory Launchpad Program
Track	CEDA Track, Data Engineering
Training Organization	PwC Acceleration Centers in India
Training Mode	Online, Self-Paced Learning
Training Period	February 2026 to July 2026
Course Level	Industry-Oriented Skill Development
Course Status	Successfully Completed

Description

The PwC Advisory Launchpad Program was completed through the PwC Launchpad learning platform as a structured industry-oriented training programme. The CEDA Data Engineering track covered programming fundamentals, data engineering, modern data systems, data analytics and emerging technologies.

The programme included practical programming exercises, quizzes, assessments and learning activities. The learning approach allowed me to gradually move from programming fundamentals to data manipulation, analytics, databases and modern AI concepts.

The training dashboard recorded **100% completion**, with **3,931 XP and Level 10**. The programme also included a leaderboard system through which learner progress could be tracked.

The training period extended from **February 2026 to July 2026**, during which I completed the assigned learning activities and assessments.

TRAINING PROGRESS

The training was completed through a self-paced learning platform and covered the following major phases:

Phase	Key Topics
1. Soft Skills	Communication, active listening, presentation, time management
2. Python Programming	Python fundamentals, OOP, functions, file handling, exception handling
3. Data Engineering	Pandas, data cleaning, EDA, statistics, visualization
4. Modern Data Systems	SQL, CRUD, indexing, NoSQL and MongoDB
5. Generative AI	GenAI fundamentals, Transformers, attention mechanisms, prompt engineering
6. AI & Applications	AI agents, GenAI tools and industry use cases

Summary of Progress

The training followed a gradual progression from foundational skills toward more specialized technology areas. Programming fundamentals provided the base for working with data, while the data engineering modules introduced practical techniques for processing and analysing information.

The modern data systems modules then expanded this understanding to databases and NoSQL systems. The final technical component introduced Generative AI, prompt engineering and AI agents, providing exposure to emerging technologies.

LEARNING OUTCOMES

The PwC Advisory Launchpad training provided me with both technical knowledge and practical exposure. The major learning outcomes are divided into technical and professional learning.

1. Technical Learning

- Developed a stronger understanding of Python programming fundamentals.
- Practiced control structures, collections, functions and modules through programming exercises.
- Gained practical experience with file handling, classes, objects, inheritance and polymorphism.
- Learned to handle exceptions and work with regular expressions.
- Developed familiarity with Pandas for data manipulation and analysis.
- Practiced indexing, selecting, row and column operations and missing-value handling.
- Learned the basics of exploratory data analysis and statistical analysis.
- Practiced univariate, bivariate and multivariate analysis.
- Worked with correlation matrices and data visualization.
- Gained exposure to web scraping and data retrieval.
- Practiced SQL operations including DDL, DML, SELECT and CRUD operations.
- Learned basic NoSQL and MongoDB operations.
- Gained exposure to Generative AI, generative models, transformers and attention mechanisms.
- Practiced basic prompt engineering through practical prompt-based activities.
- Developed introductory understanding of AI agents and their applications.

2. Professional Skills

- Improved communication and active listening skills.
- Developed better presentation and questioning skills.
- Learned techniques for managing time and working effectively in remote environments.
- Improved the ability to learn independently through self-paced education.
- Developed a more structured approach to solving technical problems.

Overall Learning

The training helped me develop a broader understanding of modern technology. Rather than focusing on a single programming language or tool, the programme exposed me to the complete progression from programming fundamentals to data processing, analytics, databases and emerging AI technologies.

SKILLS ACQUIRED

The practical training on **"Advisory Launchpad Program"** helped me develop a combination of technical and professional skills that are essential for software development and problem-solving. The key skills acquired during the training are summarized below.

Skill Area	Key Skills Developed
Python Programming	Python fundamentals, OOP, functions, file handling and exception handling
Data Engineering	Pandas, data processing, EDA and data visualization
Databases	SQL, CRUD operations, indexing, NoSQL and MongoDB
Data Analytics	Statistics, correlation, data analysis and reporting
Generative AI	Generative models, Transformers, prompt engineering and AI agents
Professional Skills	Communication, presentation, active listening and time management

Overall Skill Development

The training strengthened my technical foundation in **programming, data engineering, analytics, databases and Generative AI**, while also improving my communication and self-learning abilities.

CHALLENGES FACED DURING TRAINING

1. **Understanding Advanced Python Concepts**
   Concepts such as OOP, inheritance and polymorphism required additional practice.

2. **Working with Data**
   Data cleaning, missing values and exploratory analysis required careful understanding and practice.

3. **Learning Database Concepts**
   Understanding SQL operations and the differences between relational and NoSQL databases required hands-on practice.

4. **Understanding Generative AI**
   Concepts such as Transformers, attention mechanisms and AI agents were new and required additional study.

5. **Managing Multiple Learning Areas**
   Balancing programming, data, databases, GenAI and soft skills required consistent time management.

How I Overcame Them

Regular practice, revisiting learning material and solving practical exercises helped me gradually understand difficult concepts and become more confident in applying them.

REAL-WORLD APPLICATIONS OF LEARNING

The concepts covered during the PwC Advisory Launchpad programme have applications across modern software, data and technology environments.

1. Data Analytics

The Pandas, EDA and statistical analysis concepts can be used to process datasets, identify trends, understand patterns and support data-driven decision making. For example, businesses can analyse customer, sales or operational data to identify patterns and make better decisions.

2. Database Systems

SQL, CRUD operations, indexing and database concepts are commonly used in applications that store and retrieve structured information. The MongoDB exercises also provided exposure to document-oriented databases, which are useful in applications where flexible data structures are required.

3. Business Reporting

The analytical reporting exercises involving ranking, running totals, monthly and yearly reports and OLAP concepts demonstrate how data can be transformed into information useful for business analysis.

4. Machine Learning

The exercises involving Logistic Regression and Decision Trees provided an introduction to applying machine learning algorithms to prediction problems. These concepts can be extended to areas such as customer prediction, classification and business analytics.

5. Generative AI

The Generative AI module introduced technologies that are increasingly being used for content generation, information processing, productivity tools and industry-specific applications. Prompt engineering can help users communicate more effectively with AI systems and obtain useful outputs for specific tasks.

6. Professional Communication

The communication, presentation, active listening and time management modules are relevant to software teams, consulting environments and professional interactions where technical knowledge must be communicated clearly.

CONCLUSION

The **PwC Advisory Launchpad Program, CEDA Track** provided me with a valuable opportunity to strengthen my technical foundation and gain exposure to several areas that are relevant to the modern technology industry.

The training started with programming fundamentals using Python and gradually introduced more specialized areas such as data engineering, data analytics, databases, NoSQL systems and Generative AI. The practical exercises helped me understand how theoretical concepts can be applied to solve structured problems.

One of the most useful aspects of the programme was the combination of different technical areas. Learning Python provided a programming foundation, while Pandas and data analytics introduced methods for working with data. SQL, databases and MongoDB helped me understand data storage and retrieval, while the Generative AI modules introduced me to emerging technologies such as transformers, prompt engineering and AI agents.

The training also included professional development topics such as communication, active listening, presentation skills, time management and remote working. These modules helped me understand that technical skills need to be supported by effective communication and professional discipline.

The assessment and progress system encouraged consistent participation throughout the programme. I completed the learning path with **100% completion**, reached **Level 10**, and earned **3,931 XP** on the training platform.

Overall, the programme helped bridge the gap between academic learning and industry-oriented technical skills. It improved my confidence in working with programming, data and emerging technologies and provided a broader foundation for future projects, internships and professional opportunities.

KEY TAKEAWAYS

The PwC Advisory Launchpad Program was a valuable learning experience that helped me develop a broader understanding of modern technology and its practical applications.

One of the most important takeaways from the training was the importance of building strong fundamentals before moving toward advanced technologies. The Python programming modules provided a foundation that made it easier to understand later topics involving data processing and analytics.

The Data Engineering and Modern Data Systems modules helped me understand how data can be collected, processed, stored and analysed. Working with Pandas, SQL, MongoDB and analytical exercises gave me practical exposure to different stages of working with data.

The Generative AI module was another important part of the training. Learning about generative models, transformers, attention mechanisms, prompt engineering and AI agents helped me understand the direction in which modern AI applications are developing.

The training also reinforced the importance of continuous practice. Completing programming exercises and assessments showed me that practical implementation is important for converting theoretical knowledge into usable skills.

Another important lesson was the value of self-learning. Since the programme was largely self-paced, I had to manage my time, track my progress and complete the learning activities consistently.

Overall, the training has strengthened my foundation in **programming, data engineering, analytics, modern data systems and Generative AI**. It has also improved my problem-solving, communication and self-learning abilities.